THE SERRE TENSOR IDEAL CUT OUT BY A HOMOLOGICAL FUNCTOR, AND THE LOCALISATION DIAGRAM IT PRODUCES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned publisher source sources/jep-2020-135-homological-support/full-original-native.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement includes the Xy-pic display of the four-square diagram, and the proof's first sentence notes that the left-hand square is a repetition for the reader's convenience, which is kept verbatim.

Editorial note. Every \Cref outside the counted range is recorded as a navigation stub whose displayed number is the author's own numbering computed from the author's \newtheorem declarations; the load-bearing steps of the proof stay inside the counted spans.

Editorial note. Assembly history: the first attempt (r001) was returned with 'invalid external label' because the counted proof refers to the author's labels Prop:A/C and it:A/C-d, which contain a slash, and the assembler's external-label whitelist did not admit that character. The Chief fixed the tool on 2026-10-12 (the whitelist now admits '/', while characters that could break out of the label such as braces, backslash, hash, percent and tilde stay refused), so the card is reinstated unchanged and re-assembled as r002; no statement, span or stub changed. The withdrawn job directory was moved back into jobs/author-material-new/ on the local mirror before this run.

Allowed standard prerequisites (editorial).

- The standing hypothesis on a big tensor-triangulated category and its tensor-triangulated subcategory of compact objects. Same source, Hypothesis labelled Hyp:big-tt (source lines 180-183); invoked in the counted statement. Fixes the ambient category mT and the class of objects the construction applies to.

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- The homological functor criterion producing a maximal Serre tensor ideal. Same source, Proposition labelled Prop:Finv; invoked inside the counted proof at line 799. Supplies the maximal Serre tensor ideal cB = Ker(hat H) cap A^fp from the homological functor H = Q_cC o hat F o yoneda_T.
- Localisation of a tensor-triangulated category at a Serre tensor ideal. Same source, the notation AB = mT/Loc(cB) set up before the statement; used at lines 797-806. Produces the quotient category and the localisation functor Q_cB through which the homological functor factors.

Source reference. Original lemma 6: sources/jep-2020-135-homological-support/full-original-native.tex, source lines 784–784. Navigation pointer for \ref{Cons:E_C} used inside the counted span (line 784); the argument that consumes it is retained in the span.

Source reference. Original lemma 4: sources/jep-2020-135-homological-support/full-original-native.tex, source lines 814–814. Navigation pointer for \ref{Prop:A/C} used inside the counted span (line 814); the argument that consumes it is retained in the span.

Source reference. Original lemma 3: sources/jep-2020-135-homological-support/full-original-native.tex, source lines 799–799. Navigation pointer for \ref{Prop:Finv} used inside the counted span (line 799); the argument that consumes it is retained in the span.

Source reference. Original lemma 5: sources/jep-2020-135-homological-support/full-original-native.tex, source lines 814–814. Navigation pointer for \ref{Prop:ring-in-AB} used inside the counted span (line 814); the argument that consumes it is retained in the span.

Source reference. Original lemma 1: sources/jep-2020-135-homological-support/full-original-native.tex, source lines 767–767. Navigation pointer for \ref{Rec:T-S} used inside the counted span (line 767); the argument that consumes it is retained in the span.

Source reference. Original equation 1: sources/jep-2020-135-homological-support/full-original-native.tex, source lines 788–788. Navigation pointer for \ref{eq:hatF-hatU} used inside the counted span (line 788); the argument that consumes it is retained in the span.

Source reference. Original lemma 4: sources/jep-2020-135-homological-support/full-original-native.tex, source lines 814–814. Navigation pointer for \ref{it:A/C-d} used inside the counted span (line 814); the argument that consumes it is retained in the span.

0.1. **Lemma.** *With notation as in Original lemma* 1*, let* $\mathcal{C} \in \mathrm{Spc}^{\mathrm{h}}(\mathcal{S}^c)$*. Then* $\mathcal{B} := \big\{ M \in \text{mod-}\mathcal{T}^c \,\big|\, \hat{F}(M) \in \mathcal{C} \big\}$ *is a maximal Serre* $\otimes$*-ideal of* $\text{mod-}\mathcal{T}^c$*. Furthermore, we have a diagram*

(0.2)
$$\begin{array}{ccccc}
\mathcal{T} & \xrightarrow{\mathbb{h}_{\mathcal{T}}} & \mathcal{A} = \text{Mod-}\mathcal{T}^c & \underset{Q_{\mathcal{B}}}{\overset{R_{\mathcal{B}}}{\rightleftarrows}} & \bar{\mathcal{A}}_{\mathcal{B}} = \text{Mod-}\mathcal{T}^c/\operatorname{Loc}(\mathcal{B}) \\
F \big\downarrow \big\uparrow U & & \hat{F} \big\downarrow \big\uparrow \hat{U} & & \bar{F} \big\downarrow \big\uparrow \bar{U} \\
\mathcal{S} & \xrightarrow{\mathbb{h}_{\mathcal{S}}} & \text{Mod-}\mathcal{S}^c & \underset{Q_{\mathcal{C}}}{\overset{R_{\mathcal{C}}}{\rightleftarrows}} & \text{Mod-}\mathcal{S}^c/\operatorname{Loc}(\mathcal{C})
\end{array}$$

with $\mathbb{h}_{\mathcal{S}}\, F \cong \hat{F}\, \mathbb{h}_{\mathcal{T}}$ *and* $\mathbb{h}_{\mathcal{T}}\, U \cong \hat{U}\, \mathbb{h}_{\mathcal{S}}$ *and* $Q_{\mathcal{C}}\, \hat{F} \cong \bar{F}\, Q_{\mathcal{B}}$ *and* $\hat{U}\, R_{\mathcal{C}} \cong R_{\mathcal{B}}\, \bar{U}$*. Finally, the pure-injective* $E_{\mathcal{B}}$ *is a direct summand of* $U(E_{\mathcal{C}})$ *in* $\mathcal{T}$ *(see Original lemma* 6*).*

Proof. The left-hand square of (0.2) is (1), repeated for cognitive help. Since $\hat{F}$ is exact, $\hat{U}$ preserves injectives hence $U(E_{\mathcal{C}})$ is pure-injective. Since U is lax-monoidal, $U(E_{\mathcal{C}})$ is a weak ring, with unit $\mathbb{1} \to U(E_{\mathcal{C}})$ adjoint to $\eta_{\mathcal{C}} \colon \mathbb{1}_{\mathcal{S}} = F(\mathbb{1}_{\mathcal{T}}) \to E_{\mathcal{C}}$. In particular, $U(E_{\mathcal{C}}) \neq 0$. The internal-hom version of the adjunction reads

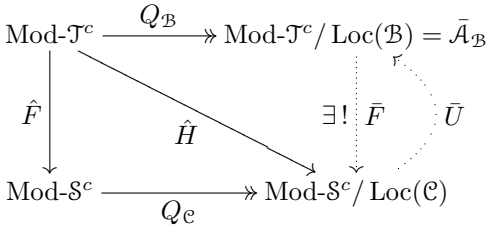
$$[M, \hat{U}(\hat{E}_{\mathcal{C}})] \cong \hat{U}[\hat{F}(M), \hat{E}_{\mathcal{C}}] \tag{0.3}$$

for every $M \in \mathcal{A}$, as can be checked by testing under $\mathrm{Hom}_{\mathcal{A}}(\hat{x}, -)$ for $x \in \mathcal{T}^c$.

Consider now the functor $H := Q_{\mathcal{C}} \circ \hat{F} \circ \mathbb{h}_{\mathcal{T}}$

$$H \colon \mathcal{T} \to \text{Mod-}\mathcal{T}^c \to \text{Mod-}\mathcal{S}^c \twoheadrightarrow \text{Mod-}\mathcal{S}^c/\operatorname{Loc}(\mathcal{C}) =: \mathcal{D}.$$

This is a homological functor to which we can apply Original lemma 3, with $\hat{H}$ being necessarily $Q_{\mathcal{C}} \circ \hat{F}$. It tells us that $\mathcal{B} = \mathrm{Ker}(\hat{H}) \cap \mathcal{A}^{\mathrm{fp}}$ is indeed a maximal Serre $\otimes$-ideal of $\mathcal{A}^{\mathrm{fp}}$. We can then factor $\hat{H}$ via $Q_{\mathcal{B}}$, yielding a unique tensor-exact coproduct-preserving functor $\bar{F} \colon \text{Mod-}\mathcal{T}^c/\operatorname{Loc}(\mathcal{B}) \to \text{Mod-}\mathcal{S}^c/\operatorname{Loc}(\mathcal{C})$ making the following diagram commute



By general Grothendieck-category theory, $\bar{F}$ admits a right adjoint $\bar{U}$. This gives us (0.2). Note that $R_{\mathcal{B}}\bar{U}$ is the right adjoint to $\bar{F}\, Q_{\mathcal{B}} \cong Q_{\mathcal{C}}\, \hat{F}$, whose right adjoint is also $\hat{U}\, R_{\mathcal{C}}$. Hence we have

$$R_{\mathcal{B}} \circ \bar{U} \cong \hat{U} \circ R_{\mathcal{C}}. \tag{0.4}$$

Consider $\bar{\mathbb{1}} \rightarrowtail \bar{E}_{\mathcal{C}}$ the injective hull of the unit in $\text{Mod-}\mathcal{S}^c/\operatorname{Loc}(\mathcal{C})$, as in the statement. The pure-injective $E_{\mathcal{C}} \in \mathcal{S}$ corresponding to $\mathcal{C}$ is then characterized by $\hat{E}_{\mathcal{C}} = R_{\mathcal{C}}(\bar{E}_{\mathcal{C}})$. Therefore, by (0.4), we have $R_{\mathcal{B}}(\bar{U}(\bar{E}_{\mathcal{C}})) \cong \hat{U}(\hat{E}_{\mathcal{C}}) = \mathbb{h}_{\mathcal{T}}(U(E_{\mathcal{C}}))$. Applying $Q_{\mathcal{B}}$ to this relation, we see that the injective $\bar{U}(\bar{E}_{\mathcal{C}})$ is the image in $\bar{\mathcal{A}}_{\mathcal{B}}$ of $U(E_{\mathcal{C}})$. By Original lemma 5 the unit of this non-trivial weak ring $\bar{U}(\bar{E}_{\mathcal{C}})$ is a monomorphism $\bar{\mathbb{1}} \rightarrowtail \bar{U}(\bar{E}_{\mathcal{C}})$ in $\bar{\mathcal{A}}_{\mathcal{B}}$ and $\bar{U}(\bar{E}_{\mathcal{C}})$ is injective. Therefore the injective hull $\bar{E}_{\mathcal{B}}$ of $\bar{\mathbb{1}}$ is a direct summand of $\bar{U}(\bar{E}_{\mathcal{C}})$. As usual, this decomposition holds in $\mathcal{T}$ already, by Original lemma 4 (4). □

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